

Chapter 1 : Important Safety Instructions

1.1 Installation Warnings

- The Active Power Filter (APF) is designed for multiple applications. It shall be connected to a power grid system, in parallel with harmonic nonlinear loads, as a solution for harmonics mitigation.
- Install the APF in a well-ventilated indoor area, away from excess moisture, heat, dust, flammable gas or explosives. To avoid fire accidents and electric shock, the indoor area must be free of conductive contaminants. For the temperature and humidity specifications, please refer to Appendix 1: Technical Specifications.
- The area around the APF must be kept clear of objects to allow ventilation and easy access for operating the machine.
- The APF shall not be exposed to the environment of corrosive gases/ particles.
- The APF shall be installed in a cabinet with a protective marking. Appropriate ventilation channels for heat dissipation shall be maintained.
- To minimize fire and electric shock hazards, installation must be conducted by qualified personnel in a controllable working environment.
- To minimize electric shock hazards, all maintenance work must be carried out by a qualified technician, and be sure to cut off all power supply before maintenance.
- After the system power is cut off, please wait for the operating switch to cool down before you perform any operation to prevent burn injury caused by the high temperature.
- High voltage hazards! It takes at least 15 minutes for the DC capacitor to discharge. Please make sure the device has discharged completely before carrying out any operation.
- To minimize electric shock hazards, please read this Manual carefully before switching the power on, and keep this Manual properly for permanent reference.

1.2 Connection Warnings

- To prevent a possible risk of current leakage, the APF shall be earthed properly.
- With regard to wiring, the compensation capacity and the current-carrying capacities of cables shall be taken into account.
- The APF's power wires must be connected to a protective device. It is suggested that every module shall be equipped with an overcurrent protection device certified by a third party. The installation location shall be taken into consideration. Please choose a protection device that has enough breaking capacity.
- The capacity of the protective devices shall fit that of the APF. Please follow the chart below to select a proper protection device in accordance with the APF capacity.